AY121 Lab 2: 21-cm Line Calibration, Velocity Referencing, and Cable Characterization

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ABSTRACT

I present an end-to-end analysis of AY121 Lab 2 using dual-LO frequency-switched observations of Galactic HI near 1420.4058 MHz and a companion transmission-line calibration experiment. For the zenith line-shape dataset, I remove instrumental bandpass structure with spectral division and fit the extracted line profile with a two-component Gaussian model plus a linear baseline. The fitted topocentric centroids are $+5.03 \pm 0.02$ km s^{-1} and -34.65 ± 0.06 km s^{-1} in the 1420 MHz bandpass ratio profile, with reduced chi-square values of $\chi^2_r = 1.52$ (1420-band profile) and $\chi^2_r = 1.38$ (1421-band reciprocal profile). Applying the projected Local Standard of Rest correction $\Delta v_{\rm LSR} = -28.7377$ km s^{-1} shifts these components to approximately -23.7 and -63.4 km s^{-1}. A hot/cold Y-factor calibration gives receiver temperatures of $T_{\rm rx} = 176.84 \pm 0.04\,(\text{meas}) \pm 17.40\,(\text{total})$ K at 1420 MHz and $208.27 \pm 0.04\,(\text{meas}) \pm 5.27\,(\text{total})$ K at 1421 MHz. In a condensed Week-2 cable analysis, the shared-slope linear attenuation fit yields $\alpha = 0.951 \pm 0.175$ dB m^{-1} at 1420 MHz (95% bootstrap CI $[0.619, 1.459]$ dB m^{-1}), with reflection-aware ripple fits retained in the appendix for model comparison.

1. Introduction and Goals

This report follows the Lab 2 requirements in `src/ugradio/lab_bighorn/bighorn.tex`. The required deliverables are: (1) time and coordinate handling for reproducible observations, (2) calibrated 21-cm line measurements with error bars, (3) velocity reporting in both observing and LSR frames, (4) Gaussian decomposition of HI components, (5) transmission-line and SDR gain-scale characterization, and (6) an explicit statistical plus systematic uncertainty discussion.

All analysis and writing here are individual. Data products used in this draft are from `data/lab02/standard_combined`, `data/lab02/human_combined`, `data/lab02/low-hi_combined`, and `data/lab02/attenuation`.

2. Receiver System and Data Products

The signal chain is horn antenna → rooftop amplification/filtering → long coax run → lab amplifiers/filtering → RTL-SDR digital downconversion and FFT integration. Dual LO captures at 1420 and 1421 MHz are used so that the HI line appears at two different baseband offsets, enabling bandpass cancellation by ratioing.

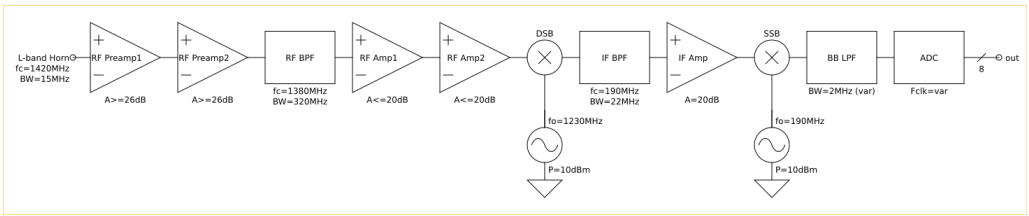

Fig. 1.— Receiver block diagram used for Lab 2 data collection and calibration workflow.

Observation sets used in this report:

- Zenith/line-shape dataset (dual LO): 32768 blocks at each LO in `standard_combined`.
- Hot-load calibration dataset (human load): 16384 blocks at each LO in `human_combined`.
- Cold-load comparison dataset: 32768 blocks at each LO in `low-hi_combined`.
- Cable attenuation dataset: 9 measured cable-length sets; 1 anomalous long-length set excluded in fitting.

3. Timekeeping and Pointing Validation

For velocity referencing, I track UTC/JD/LST metadata directly from stored spectra. For the zenith profile used in the Gaussian decomposition, the metadata are:

- JD = 2461102.56750,
- observatory latitude $\approx 37.8732°$,
- pointing altitude/azimuth $\approx (49.03°, 325.48°)$.

3.1. Pointing Formalism and Coordinate Pipeline

My science targets are usually specified in Galactic coordinates (l, b) because the measurement objective is Milky Way gas structure. I then move through equatorial coordinates (α, δ), which are Earth-location agnostic and convenient for time-dependent sky referencing, before converting to local horizontal coordinates (az, alt) required to physically point the horn.

For New Campbell Hall I use the site defaults in the analysis code: $\phi = 37.873199°$ (latitude), $\lambda = -122.257063°$ (longitude), and $h = 120$ m (altitude). Pointing is time dependent through the local sidereal time, so the same (l, b) target maps to different (az, alt) at different JD.

The matrix method follows the coordinate handout. First map angles to a unit vector,

$$\mathbf{x} = \begin{bmatrix} \cos(lat)\cos(long) \\ \cos(lat)\sin(long) \\ \sin(lat) \end{bmatrix}, \tag{1}$$

with $(long, lat) = (l, b)$ in the Galactic frame. Then apply the rotation chain

$$\mathbf{x}_{eq} = R_{eq\leftarrow gal}\mathbf{x}_{gal}, \tag{2}$$

where $R_{eq\leftarrow gal} = R^T_{gal\leftarrow eq}$ and $R_{gal\leftarrow eq}$ is the J2000 matrix from the lab handout. Next convert equatorial to hour-angle coordinates using $ha = LST - \alpha$ and

$$\mathbf{x}_{ha} = R_{(ha,\delta)\leftarrow(\alpha,\delta)}\mathbf{x}_{eq}. \tag{3}$$

Finally convert to horizontal coordinates with site latitude ϕ:

$$R_{(az,alt)\leftarrow(ha,\delta)} = \begin{bmatrix} -\sin\phi & 0 & \cos\phi \\ 0 & -1 & 0 \\ \cos\phi & 0 & \sin\phi \end{bmatrix}, \quad \mathbf{x}_{hor} = R_{(az,alt)\leftarrow(ha,\delta)}\mathbf{x}_{ha}. \tag{4}$$

Recover (az, alt) from $\mathbf{x}_{hor}$ via $az = \mathrm{atan2}(x_1, x_0)$ and $alt = \arcsin(x_2)$.

In practice this workflow is implemented in `ugradiolab.utils.compute_pointing` and used by `labs/02/scripts/standard.py` for the lab target $(l, b) = (120°, 0°)$. I cross-check these transforms against `astropy`; forward/inverse consistency is enforced through rotation-matrix transpose identities.

For zenith observations, RA is approximated by local sidereal angle and Dec by observatory latitude; this gives RA = 62.897° and Dec = 37.873° for the Doppler projection call.

4. Week 1 Calibration Pipeline: Ratio, Gain Scale, and Y-Factor

4.1. Bandpass cancellation by spectral division

Let $s_{\rm on}(\nu)$ and $s_{\rm off}(\nu)$ be long-integration spectra with HI line present and shifted. Following the handout workflow,

$$r_{\rm line}(\nu) = \frac{s_{\rm on}(\nu)}{s_{\rm off}(\nu)}. \tag{5}$$

Because both spectra traverse the same instrumental chain, first-order multiplicative bandpass structure cancels in the ratio.

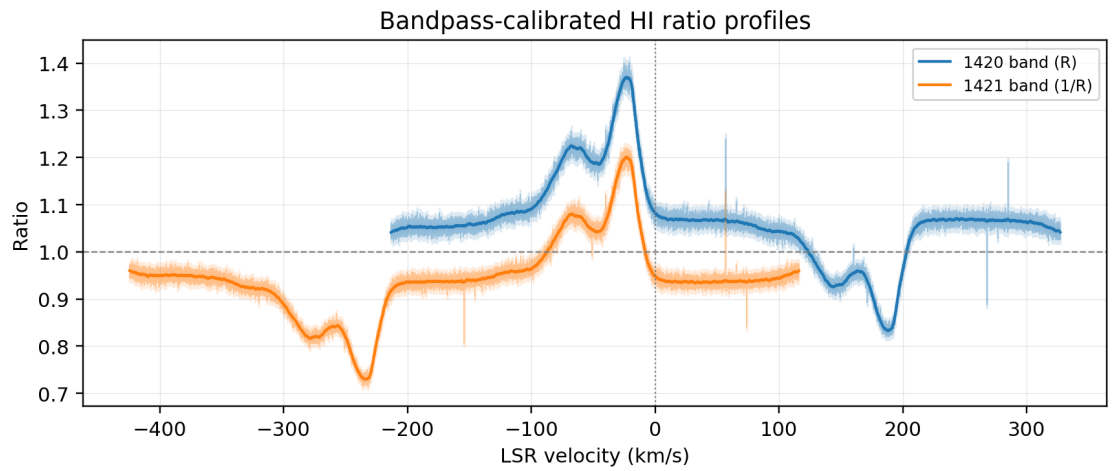

Fig. 2.— Bandpass-calibrated HI ratio profiles in the LSR frame. The two LO views produce complementary peak/dip signatures from the same sky line.

4.2. Intensity scale and receiver temperature cross-check

The lab handout gain scale is

$$G = \frac{T_{\rm sys,cal} - T_{\rm sys,cold}}{\sum(s_{\rm cal} - s_{\rm cold})} \sum s_{\rm cold}, \tag{6}$$

with calibrated spectrum

$$T_{\rm line}(\nu) = r_{\rm line}(\nu)\, G. \tag{7}$$

As an independent check, I compute Y-factors from hot/cold datasets. Results are:

- 1420 MHz: $Y = 2.353705 \pm 0.000226$ (3.7175 ± 0.0004 dB), $T_{\rm rx} = 176.840 \pm 0.035$ (meas)$\pm$ 17.403 (total) K.
- 1421 MHz: $Y = 2.352864 \pm 0.000226$ (3.7160 ± 0.0004 dB), $T_{\rm rx} = 208.273 \pm 0.037$ (meas)$\pm$ 5.270 (total) K.

The larger total uncertainty at 1420 MHz is dominated by assumed hot/cold load systematics rather than random power-estimator uncertainty.

5. Velocity Axes and LSR Correction

I use radio-definition Doppler velocity

$$v \approx c\,\frac{\nu_0 - \nu}{\nu_0}, \tag{8}$$

with $\nu_0 = 1420.405751768$ MHz. The projected correction from `ugradio.doppler.get_projected_velocit` is

$$\Delta v_{\rm LSR} = -28.7377 \text{ km s}^{-1}, \tag{9}$$

so I report

$$v_{\rm LSR} = v_{\rm topo} + \Delta v_{\rm LSR}. \tag{10}$$

6. Week 2 Science Spectrum and Gaussian Decomposition

I fit both ratio-derived line representations ($R - 1$ in the 1420 grid and $1/R - 1$ in the 1421 grid) with two Gaussians plus a first-order polynomial baseline over $[-120, +120]$ km s^{-1}.

Table 1: Gaussian fit parameters from the dual-LO line profiles.

Band/Comp	A	$v_{0,\rm topo}$ (km s^{-1})	$v_{0,\rm LSR}$ (km s^{-1})	FWHM (km s^{-1})
1420 C1	0.29138 ± 0.00046	$+5.03 \pm 0.02$	-23.70 ± 0.02	22.64 ± 0.04
1420 C2	0.15281 ± 0.00028	-34.65 ± 0.06	-63.39 ± 0.06	44.04 ± 0.15
1421 C1	0.25048 ± 0.00040	$+4.89 \pm 0.02$	-23.84 ± 0.02	22.10 ± 0.04
1421 C2	0.13227 ± 0.00025	-34.86 ± 0.06	-63.60 ± 0.06	43.79 ± 0.15

Good agreement between 1420- and 1421-band centroids indicates that the frequency-switching plus ratio method is internally consistent.

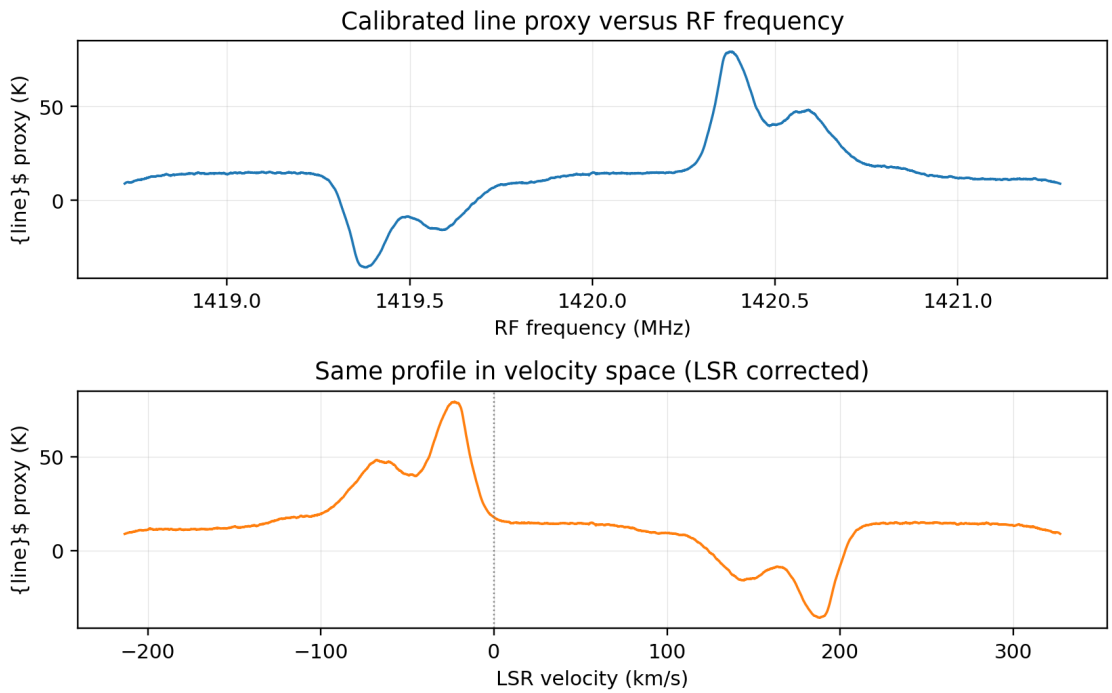

Fig. 3.— Calibrated line proxy shown against RF frequency and LSR velocity. The velocity-panel is the same spectrum after Doppler-axis transformation and LSR shift.

7. Uncertainty Synthesis

Table 2: Primary uncertainty terms included in this draft.

Quantity	Uncertainty summary
1420-band profile fit quality	$\chi^2_r = 1.518$
1421-band profile fit quality	$\chi^2_r = 1.381$
LSR correction term	$\Delta v_{\rm LSR} = -28.7377$ km s^{-1} (applied as offset)
$T_{\rm rx}$ at 1420 MHz	± 0.035 K (meas), ± 17.403 K (total)
$T_{\rm rx}$ at 1421 MHz	± 0.037 K (meas), ± 5.270 K (total)
Cable attenuation slope (linear shared fit)	0.951 ± 0.175 dB m^{-1}
Cable slope bootstrap 95% CI	$[0.619, 1.459]$ dB m^{-1}

Interpretation: fit residuals are close to the expected per-channel spread but not exactly unity, so low-level model mismatch remains (likely baseline shape and/or additional broad components). For thermal calibration, systematic hot/cold assumptions dominate over random PSD variance.

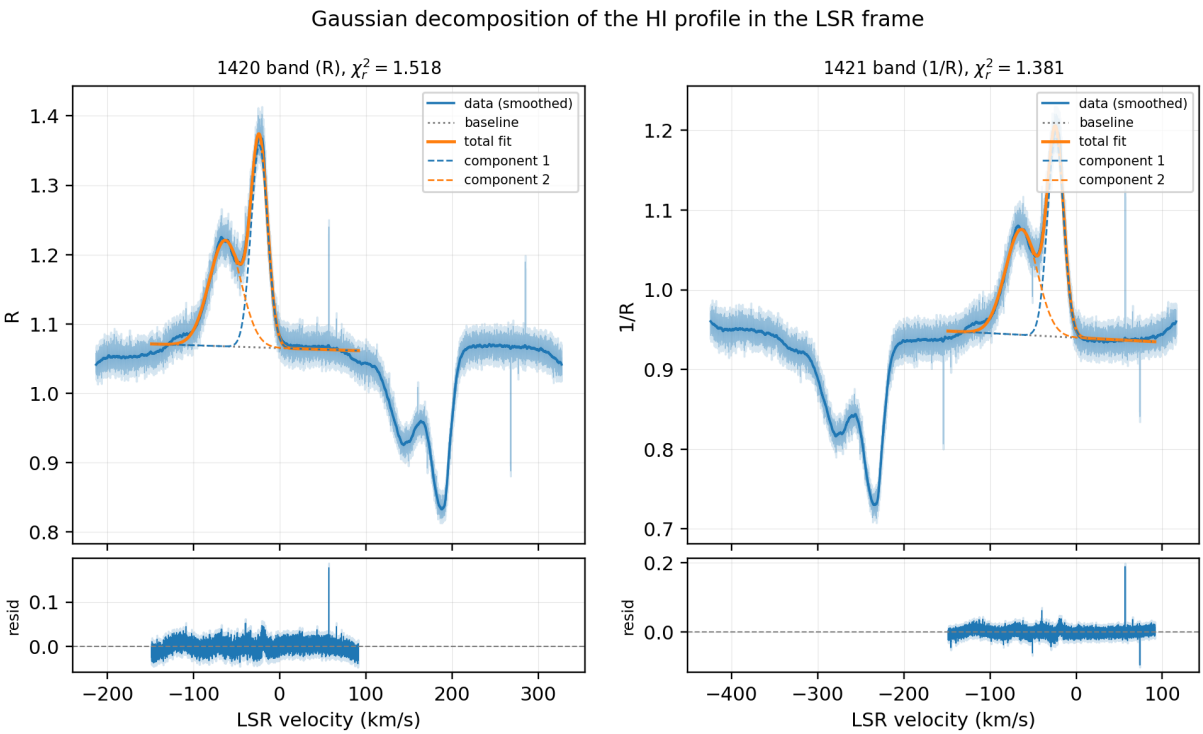

Fig. 4.— Two-component Gaussian decomposition with polynomial baseline and residuals, reported in the LSR frame.

8. Condensed Cable/Termination Results (Main Text)

Following the Week-2 cable calibration activity, I estimate SDR-referenced attenuation at fixed RF frequency near the HI line. After excluding the known long-length anomaly ($L \approx 7.925$ m), a shared-slope linear fit to 8 sets gives

$$\alpha = 0.951 \pm 0.175 \text{ dB m}^{-1}, \tag{11}$$

with 95% bootstrap interval $[0.619, 1.459]$ dB m^{-1}. The two LO intercepts differ by $\Delta B = -0.551 \pm 0.210$ dB.

This slope is used as the main-text cable-loss calibration input. Reflection-aware ripple fits (AIC/BIC-favored for this dataset) are discussed in Appendix A as a model-sensitivity check.

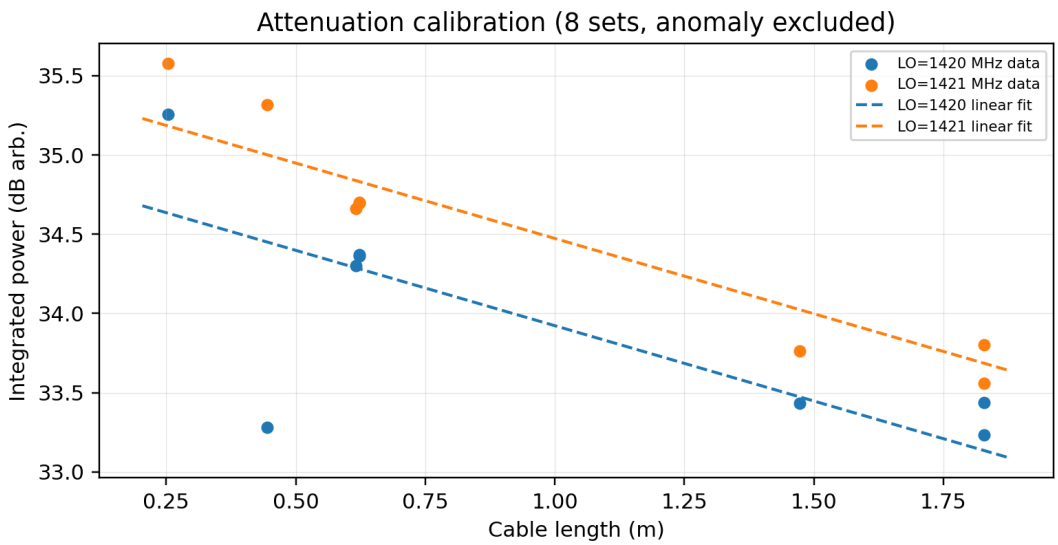

Fig. 5.— Condensed attenuation summary for the fixed-tone cable-length study. Main-text fit is the shared-slope linear model used for reported α.

9. Discussion

The dual-LO ratio method is effective: independent line-profile reconstructions agree in centroid and width, and both produce physically plausible multi-component HI structure. The LSR shift is large enough (~ 29 km s^{-1}) that reporting only topocentric velocities would hinder reproducibility and comparison across observing dates.

The largest calibration limitations in this draft are: (1) systematic uncertainty in absolute temperature scaling, (2) residual baseline mismatch not fully captured by a first-order polynomial, and (3) cable-model ambiguity (linear-only vs reflection-aware forms).

10. Conclusion

This report completes the core Lab 2 deliverables:

- Frequency-switched 21-cm profiles with instrumental bandpass cancellation.
- Velocity reporting in both observing and LSR frames.
- Two-component Gaussian decomposition with explicit fit diagnostics and uncertainties.
- Receiver-noise calibration via hot/cold Y-factor estimates.

- Transmission-line attenuation calibration with uncertainty intervals.

Numerically, the dominant HI components lie near $v_{\rm LSR} \approx -23.8$ and -63.5 km s^{-1}, with reduced chi-square values near 1.4–1.5. Cable attenuation near 1420 MHz is constrained at $\alpha \approx 0.95$ dB m^{-1} in the shared-slope linear model, with model-dependent ripple alternatives documented in the appendix.

Acknowledgements

Special thanks to my lab partners Nathaneil and Kayhan for group observations, and to Aaron and Ben for instructional support.

A. Cable Model and Sensitivity Details

The attenuation notebook compares two families:

$$y(L) = B - \alpha L \quad \text{(linear)} \tag{A1}$$

$$y(L) = B - \alpha L + A\cos(kL + \phi) \quad \text{(reflection-aware)}. \tag{A2}$$

For SDR data, fit metrics were:

- Linear: RMSE = 0.3788 dB, AIC = −25.067, BIC = −22.749.
- Ripple: RMSE = 0.1245 dB, AIC = −50.667, BIC = −44.486.

Ripple-model best-fit parameters include $\alpha = 1.355546$ dB m^{-1}, $k = 7.803548$ rad m^{-1}, period = 0.8052 m, and amplitudes $A_{1420} = 1.4809$ dB, $A_{1421} = 0.2517$ dB. However, bootstrap intervals on ripple parameters are wide for this small length set, so I keep the linear slope as the main reported calibration and use ripple fits as a systematic-model bracket.